Well-Ordering and Choice are equivalent in ZF

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/olp-set-theory-choice/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Card cut from the FREE slice of the lit pool for unit olp-set-theory-choice; cross-chapter references resolved to their pinned source units.

Editorial scope. Single theorem card cut from the Open Logic Project unit olp-set-theory-choice (pinned-original-source.tex); statement and proof are the author's own text at source lines 1202-1204 and 1206-1240.

Allowed standard prerequisites (editorial).

- General Recursion theorem (set-theory spine chapter, Theorem transrecursionschema, source line 730). cited at proof line 1226. Licenses the transfinite recursion defining the enumeration term g; the author notes g is first only a term and must be shown to be a genuine function.
- Hartogs' lemma (set-theory-choice chapter, Lemma HartogsLemma, source line 674). cited at proof line 1237. Bounds the ordinals admitting injections into a set; contradicts a never-stopping run of the recursion, forcing it to stop.

Source reference. Original theorem 0.16: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 730–730. Cited at proof line 1226: justifies the transfinite recursion defining the enumeration term g.

Source reference. Original lemma 0.7: sources/olp-set-theory-choice/pinned-original-source.tex, source lines 674–674. Cited at proof line 1237: bounds the run of the recursion, forcing it to stop.

Theorem 0.1 (in ZF)**.** *Well-Ordering and Choice are equivalent.*

Proof. Left-to-right. Let A be a set of sets. Then $\bigcup A$ exists by the Axiom of Union, and so by Well-Ordering there is some $<$ which well-orders $\bigcup A$. Now let $f(x) =$ the $<$-least member of x. This is a choice function for A.

Right-to-left. Fix A. By Choice, there is a choice function, f, for $\wp(A) \setminus \{\emptyset\}$. Using Transfinite Recursion, define a function:

$$g(0) = f(A)$$
$$g(\alpha) = \begin{cases} \text{stop!} & \text{if } A = \text{im}(g)\alpha \\ f(A \setminus \text{im}(g)\alpha) & \text{otherwise} \end{cases}$$

The indication to "stop!" is just a shorthand for what would otherwise be a more long-winded definition. That is, when $A = \text{im}(g)\alpha$ for the first time, let $g(\delta) = A$ for all $\delta \leq \alpha$. Now, in the first instance, we can only be sure that this defines a *term* (see the remarks after 0.16); but we will show that we indeed have a function.

Since f is a choice function, for each α (when defined) we have $g(\alpha) = f(A \setminus \text{im}(g)\alpha) \in A \setminus \text{im}(g)\alpha$; i.e., $g(\alpha) \notin \text{im}(g)\alpha$. So if $g(\alpha) = g(\beta)$ then $g(\beta) \notin \text{im}(g)\alpha$, i.e., $\beta \notin \alpha$, and similarly $\alpha \notin \beta$. So $\alpha = \beta$, by Trichotomy. So g is injective.

Next, observe that we do stop!, i.e. that there is some (least) ordinal α such that $A = g[\alpha]$. For suppose otherwise; then as g is injective we would have $\alpha \prec \wp(A) \setminus \{\emptyset\}$ for every ordinal α, contradicting 0.7 . Hence also $\text{ran}\, g = A$.

Assembling these facts, g is bijection from some ordinal to A. Now g can be used to well-order A. □